■ 2.2.13 Setting up Rules with -> and :>

Just as you can make both immediate and delayed definitions in *Mathematica*, you can also set up both immediate and delayed transformation rules.

lhs -> *rhs*	*rhs* is evaluated when the rule is given
lhs :> *rhs*	*rhs* is evaluated when the rule is used

Two types of transformation rules in *Mathematica*.

You should use :> whenever you need to "execute a program" to evaluate the right-hand side of the transformation rule.

```
In[1]:= t = f[(1+x)^2] + f[(1-x)^2] + g[(1+x)^2]

                  2             2             2
Out[1]= f[(1 - x) ] + f[(1 + x) ] + g[(1 + x) ]
```

This replacement executes the "program" f[Expand[x]] on each object of the form f[x_].

```
In[2]:= t /. f[x_] :> f[Expand[x]]

                         2                   2             2
Out[2]= f[1 - 2 x + x ] + f[1 + 2 x + x ] + g[(1 + x) ]
```

An immediate replacement would have performed the Expand when the replacement was first given, and in this case would not have done what you wanted.

```
In[3]:= f[x_] -> f[Expand[x]]
Out[3]= f[x_] -> f[x]
```

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